

Pregnancy Disorders: A Potential Role for Mitochondrial Altered Homeostasis.

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Pregnancy is a complex and challenging process associated with physiological changes whose objective is to adapt the maternal organism to the increasing energetic requirements due to embryo and fetal development. A failed adaptation to these demands may lead to pregnancy complications that threaten the health of both mothers and their offspring. Since mitochondria are the main organelle responsible for energy generation in the form of ATP, the adequate state of these organelles seems crucial for proper pregnancy development and healthy pregnancy outcomes. The homeostasis of these organelles depends on several aspects, including their content, biogenesis, energy production, oxidative stress, dynamics, and signaling functions, such as apoptosis, which can be modified in relation to diseases during pregnancy. The etiology of pregnancy disorders like preeclampsia, fetal growth restriction, and gestational diabetes mellitus is not yet well understood. Nevertheless, insufficient placental perfusion and oxygen transfer are characteristic of many of them, being associated with alterations in the previously cited different aspects of mitochondrial homeostasis. Therefore, and due to the capacity of these multifactorial organelles to respond to physiological and pathophysiological stimuli, it is of great importance to gather the currently available scientific information regarding the relationship between main pregnancy complications and mitochondrial alterations. According to this, the present review is intended to show clear insight into the possible implications of mitochondria in these disorders, thus providing relevant information for further investigation in relation to the investigation and management of pregnancy diseases.

Mitochondrial Dysfunction, Oxidative Stress, and Therapeutic Strategies in Diabetes, Obesity, and Cardiovascular Disease.

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Mitochondria are subcellular organelles involved in essential cellular functions, including cytosolic calcium regulation, cell apoptosis, and reactive oxygen species production. They are the site of important biochemical pathways, including the tricarboxylic acid cycle, parts of the ureagenesis cycle, or haem synthesis. Mitochondria are responsible for the majority of cellular ATP production through OXPHOS. Mitochondrial dysfunction has been associated with metabolic pathologies such as diabetes, obesity, hypertension, neurodegenerative diseases, cellular aging, and cancer. In this article, we describe the pathophysiological changes in, and mitochondrial role of, metabolic disorders (diabetes, obesity, and cardiovascular disease) and their correlation with oxidative stress. We highlight the genetic changes identified at the mtDNA level. Additionally, we selected several representative biomarkers involved in oxidative stress and summarize the progress of therapeutic strategies.

Unsymmetric hydroxylamine and hydrazine BAM15 derivatives as potent mitochondrial uncouplers.

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Chemical mitochondrial uncouplers are protonophoric, lipophilic small molecules that transport protons from the mitochondrial intermembrane space into the matrix independent of ATP synthase, thus

uncoupling nutrient oxidation from ATP production. Our previous work identified BAM15 (IC₅₀ 0.27 μ M) as a potent and efficacious mitochondrial uncoupler with potential for obesity treatment. In this paper, we investigate in vitro and in vivo properties of hydroxylamine and hydrazine BAM15 derivatives and reveal the high uncoupling nature of these compounds. Our structure-activity relationship studies revealed that the hydroxylamine BAM15 analogs are more potent than hydrazine ones. For example, the most potent of the hydrazine series was 5a with an EC₅₀ value of 4.6 μ M and 103 % activity of BAM15 while compound 4e was the best among the hydroxylamine series with EC₅₀ value of 340 nM and 118 % BAM15 mitochondrial uncoupling activity in rat L6 myoblasts. Pharmacokinetic profiling of 5a and 4e revealed low exposure (2-220 nM) and short half-life (15-27 min) in mice.

Hippo-YAP signaling alleviates copper-induced mitochondrial dysfunction and oxidative damage via the ATOX1-PPA2 pathway.

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Hippo signaling plays a crucial role in the cellular response to various stressors, such as mechanical stress, metabolic stress, and hypoxic stress. However, its physiological significance in copper (Cu) stress remains poorly understood. Here, we demonstrated aberrant activation of Hippo-YAP signaling in sheep pancreas and pancreatic organoids exposed to excessive Cu, accompanied by significant pathological changes, elevated levels of oxidative stress, and impaired mitochondrial structure and function. The inhibition of Hippo signaling or overexpression of YAP protected against Cu-induced damage by improving mitochondrial function and maintaining cellular Cu homeostasis. YAP interacted with TEAD and upregulated the expression of Cu chaperone ATOX1, a key regulator of intracellular Cu homeostasis. ATOX1 restored mitochondrial function under Cu stress by reducing mitochondrial superoxide levels, increasing ATP production and mitochondrial membrane potential. Additionally, our findings confirmed that ATOX1 indirectly bound to the PPA2 promoter and increased its transcription. Notably, the restoration of ATP production in mitochondria mediated by PPA2 overexpression facilitated efficient intracellular Cu efflux, allowing rapid and precise reestablishment of intracellular Cu homeostasis under Cu stress. Collectively, Hippo-YAP signaling alleviates Cu-induced oxidative damage by restoring mitochondrial function through activation of PPA2 transcription depending on ATOX1, thereby ensuring cellular Cu efflux and enhancing antioxidant capacity.

Cytochrome c levels link mitochondrial function to plant growth and stress responses through changes in SnRK1 pathway activity.

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Energy is required for growth as well as for multiple cellular processes. During evolution, plants developed regulatory mechanisms to adapt energy consumption to metabolic reserves and cellular needs. Reduced growth is often observed under stress, leading to a growth-stress trade-off that governs plant performance under different conditions. In this work, we report that plants with reduced levels of the mitochondrial respiratory chain component cytochrome c (CYTc), required for electron transport coupled to oxidative phosphorylation and ATP production, show impaired growth and increased global expression of stress-responsive genes, similar to those observed after inhibiting the respiratory chain or the mitochondrial ATP synthase. CYTc-deficient plants also show activation of the SnRK1 pathway, which

regulates growth, metabolism, and stress responses under carbon starvation conditions, even though their carbohydrate levels are not significantly different from wild-type. Notably, loss-of-function of the gene encoding the SnRK1 α 1 subunit restores the growth of CYTc-deficient plants, as well as autophagy, free amino acid and TOR pathway activity levels, which are affected in these plants. Moreover, increasing CYTc levels decreases SnRK1 pathway activation, reflected in reduced SnRK1 α 1 phosphorylation, with no changes in total SnRK1 α 1 protein levels. Under stress imposed by mannitol, the growth of CYTc-deficient plants is relatively less affected than that of wild-type plants, which is also related to the activation of the SnRK1 pathway. Our results indicate that SnRK1 function is affected by CYTc levels, thus providing a molecular link between mitochondrial function and plant growth under normal and stress conditions.

Targeting mitochondrial dynamics: A promising approach for intracerebral hemorrhage therapy.

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Intracerebral hemorrhage (ICH) is a major global health issue with high mortality and disability rates. Following ICH, the hematoma exerts direct pressure on brain tissue, and blood entering the brain directly damages neurons and the blood-brain barrier. Subsequently, oxidative stress, inflammatory responses, apoptosis, brain edema, excitotoxicity, iron toxicity, and metabolic dysfunction around the hematoma further exacerbate brain tissue damage, leading to secondary brain injury (SBI). Mitochondria, essential for energy production and the regulation of oxidative stress, are damaged after ICH, resulting in impaired ATP production, excessive reactive oxygen species (ROS) generation, and disrupted calcium homeostasis, all of which contribute to SBI. Therefore, a central factor in SBI is mitochondrial dysfunction. Mitochondrial dynamics regulate the shape, size, distribution, and quantity of mitochondria through fusion and fission, both of which are crucial for maintaining their function. Fusion repairs damaged mitochondria and preserves their health, while fission helps mitochondria adapt to cellular stress and removes damaged mitochondria through mitophagy. When this balance is disrupted following ICH, mitochondrial dysfunction worsens, oxidative stress and metabolic failure are exacerbated, ultimately contributing to SBI. Targeting mitochondrial dynamics offers a promising therapeutic approach to restoring mitochondrial function, reducing cellular damage, and improving recovery. This review explores the latest research on modulating mitochondrial dynamics and highlights its potential to enhance outcomes in ICH patients.

Enhancement of Mitochondrial Homeostasis: A Novel Approach to Attenuate Hypoxic Myocardial Injury.

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Cardiomyocytes are highly oxygen-dependent cells, relying on oxygen-driven oxidative phosphorylation to maintain their function. During hypoxia, mitochondrial ATP production decreases, leading to calcium overload, acidosis, and oxidative stress, which collectively trigger myocardial injury. Ischemic heart disease, caused by coronary atherosclerosis, results in myocardial ischemia and hypoxia, leading to ischemia-reperfusion (I/R) injury. Early myocardial injury is attributed to ischemia and hypoxia, but even after thrombolytic therapy, interventional surgery, or coronary artery bypass grafting (CABG) restores local blood flow and oxygen supply, myocardial reperfusion injury (I/R) may still occur. Mitochondria,

often referred to as the "powerhouses" of the cell, play a crucial role in cellular energy production. In the early stages of ischemia and hypoxia, mitochondrial dysfunction disrupts mitochondrial homeostasis, causing the accumulation of unfolded or misfolded proteins in the mitochondria. This activates the mitochondrial unfolded protein response (mtUPR) and mitophagy, which work to clear damaged proteins and mitochondria, playing a key role during this period. This review focuses on mitochondrial mechanisms during the ischemic phase of ischemia-reperfusion injury, aiming to provide new theoretical foundations and potential therapeutic strategies to reduce myocardial damage.

Mitochondrial Mutations in Cardiovascular Diseases: Preliminary Findings.

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Background/Objectives: Mitochondria are the main organelles for ATP synthesis able to produce energy for several different cellular activities. Cardiac cells require high amounts of energy and, thus, they contain a high number of mitochondria. Consequently, mitochondrial dysfunction in these cells is a crucial factor for the development of cardiovascular diseases. Mitochondria constitute central regulators of cellular metabolism and energy production, producing approximately 90% of the cells' energy needs in the form of ATP via oxidative phosphorylation. The mitochondria have their own circular, double-stranded DNA encoding 37 genes. Any mitochondrial DNA sequence anomaly may result in defective oxidative phosphorylation and lead to cardiac dysfunction. **Methods:** In this study, we investigated the potential association between mitochondrial DNA mutation and cardiovascular disease. Cardiac tissue and serum samples were collected from seven patients undergoing coronary artery bypass grafting. Total DNA was extracted from cardiac muscle tissue specimens and serum and each sample was subjected to polymerase chain reaction (PCR) to amplify the NADH dehydrogenase 1 (ND1) gene, which is part of the mitochondrial complex I enzyme complex and was screened for mutations. **Results:** We identified one patient with a homoplasmic A to G substitution mutation in cardiac tissue DNA and two patients with heteroplasmic A3397G mutation in serum DNA. Specifically, amplicon sequence analysis revealed a homoplasmic A3397G substitution in the ND1 gene in a tissue sample of the patient with ID number 1 and a heteroplasmic mutation in A3397G in serum samples of patients with ID numbers 3 and 6, respectively. The A to G substitution changes the amino acid from methionine (ATA) to valine (GTA) at position 31 of the ND1 gene. **Conclusions:** The detection of this novel mutation in patients with coronary artery disease may contribute to our understanding of the association between mitochondrial dysfunction and the disease, implying that mitochondria may be key players in the pathogenesis of cardiovascular diseases.

Mitochondrial Respiration Quantification in Yeast Whole Cells.

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Metabolism is mainly coordinated by cellular energy availability and environmental conditions. Assays for knowing how cells adapt energetic metabolism to different nutritional and environmental conditions give valuable information to elucidate molecular mechanisms. Oxidative phosphorylation is the primary source of ATP in most cells, and mitochondrial respiration activity is a key component of oxidative phosphorylation, maintaining mitochondrial membrane potential for ATP synthesis. Mitochondrial respiration is often studied in isolated mitochondria that are missing the cellular context. Here, we present a method for quantifying mitochondrial respiration in yeast-intact cells. This method applies to

any yeast species, although it has been generally used for *Saccharomyces cerevisiae* cells. First, the yeast growth in specific conditions is tested. Then, cells are washed and resuspended in deionized water with a 1:1 ratio (w/v). Cells are then placed in an oximeter chamber with constant stirring, and a Clark electrode is used to quantify oxygen consumption. Some molecules are sequentially placed into the chamber and selected according to this effect on the electron transport chain or ATP synthesis. ATPase inhibitor oligomycin is first added to measure respiration coupled to ATP synthesis. Afterward, an uncoupler is used to measure the maximal respiratory capacity. Finally, a mix of electron transport chain inhibitors is added to discard oxygen consumption unrelated to mitochondrial respiration. Data are analyzed using a linear regression to obtain the slope, representing the oxygen consumption rate. The advantage of this method is that it is specific for yeast mitochondrial respiration, maintaining the cellular context. It is essential to highlight that inhibitors used in mitochondrial respiration quantification could vary between yeast species.

Regulation of mitochondrial oxidative phosphorylation through tight control of cytochrome c oxidase in health and disease - Implications for ischemia/reperfusion injury, inflammatory diseases, diabetes, and cancer.

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Mitochondria are essential to cellular function as they generate the majority of cellular ATP, mediated through oxidative phosphorylation, which couples proton pumping of the electron transport chain (ETC) to ATP production. The ETC generates an electrochemical gradient, known as the proton motive force, consisting of the mitochondrial membrane potential ($\Delta\Psi_m$, the major component in mammals) and ΔpH across the inner mitochondrial membrane. Both ATP production and reactive oxygen species (ROS) are linked to $\Delta\Psi_m$, and it has been shown that an imbalance in $\Delta\Psi_m$ beyond the physiological optimal intermediate range results in excessive ROS production. The reaction of cytochrome c oxidase (COX) of the ETC with its small electron donor cytochrome c (Cyt c) is the proposed rate-limiting step in mammals under physiological conditions. The rate at which this redox reaction occurs controls $\Delta\Psi_m$ and thus ATP and ROS production. Multiple mechanisms are in place that regulate this reaction to meet the cell's energy demand and respond to acute stress. COX and Cyt c have been shown to be regulated by all three main mechanisms, which we discuss in detail: allosteric regulation, tissue-specific isoforms, and post-translational modifications for which we provide a comprehensive catalog and discussion of their functional role with 55 and 50 identified phosphorylation and acetylation sites on COX, respectively. Disruption of these regulatory mechanisms has been found in several common human diseases, including stroke and myocardial infarction, inflammation including sepsis, and diabetes, where changes in COX or Cyt c phosphorylation lead to mitochondrial dysfunction contributing to disease pathophysiology. Identification and subsequent targeting of the underlying signaling pathways holds clear promise for future interventions to improve human health. An example intervention is the recently discovered noninvasive COX-inhibitory infrared light therapy that holds promise to transform the current standard of clinical care in disease conditions where COX regulation has gone awry.